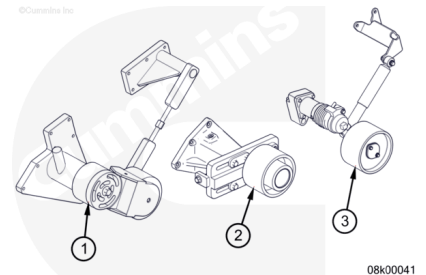


Trainings ▾

General Information

The engine can be equipped with a spring type tensioner (1), a slide type tensioner (2), or an automatic type tensioner (3).

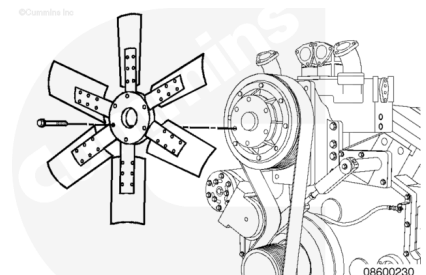


Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

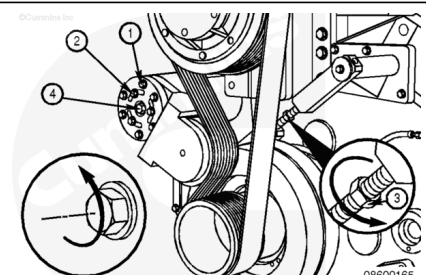
- Remove the cooling fan. Refer to Procedure 008-040 in Section 8. (</qs3/pubsys2/xml/en/procedures/89/89-008-040.html>)



Remove

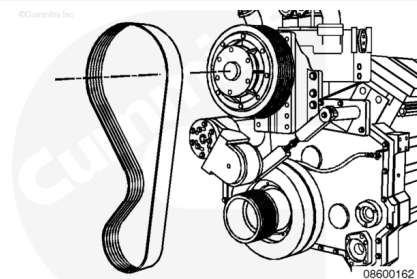
⚠ WARNING ⚠

The fan belt idler is under tension. Do not allow your hands to get between the idler and the belt or the fan hub. Personal injury can result.



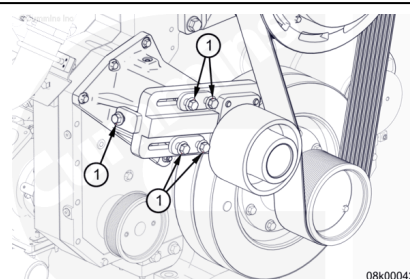
For the spring type tensioner, loosen the six capscrews (1) and the three capscrews (2) and the center capscrew (4). Turn the hexagon nut (3) **counterclockwise** to relieve tension on the belt.

Remove the fan belt.



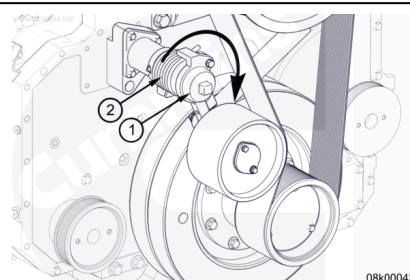
For the slide type tensioner, loosen the four mounting capscrews (1).

Turn the tensioning capscrew to relieve the tension on the belt.



For the automatic type tensioner (when removing the belt **only**):

Use a large wrench to hold the idler cap (1). Rotate the idler cap **clockwise** to relieve the tension on the idler (2).

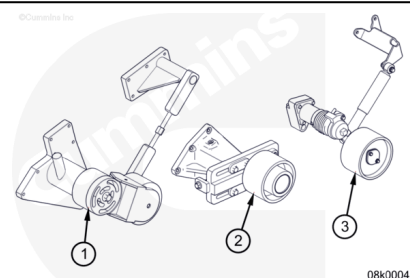


Inspect for Reuse

Inspect the spring type tensioner (1), a slide type tensioner (2), or an automatic type tensioner (3), for any cracks or excessive wear.

If the parts are cracked or worn, they **must** be replaced.

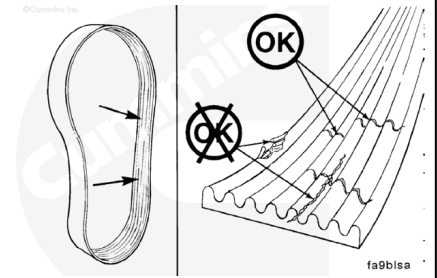
- Make sure the idler pulley moves freely
- Make sure bearing operation is smooth and does **not** have any free play
- Inspect the idler pulley for damage or wear.



Inspect the belt for:

- Cracks
- Glazing
- Tears and cuts.

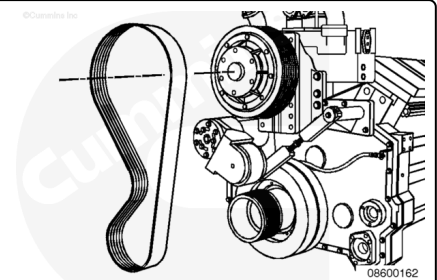
Replace the belt if damaged.



Install

For the spring type tensioner or slide type tensioner **only**:

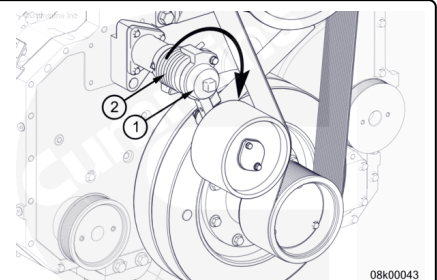
Install the belt.



For the automatic type tensioner (when fitting the belt **only**):

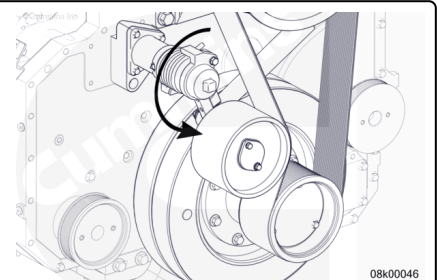
Use a large wrench to hold the idler cap (1). Rotate the idler cap **clockwise** to relieve the tension on the idler (2).

Install the belt. Make sure it is fully seated on the pulley grooves.



Slowly rotate the idler **counterclockwise** onto the belt.

Note : Belt tension is automatically set.



Adjust

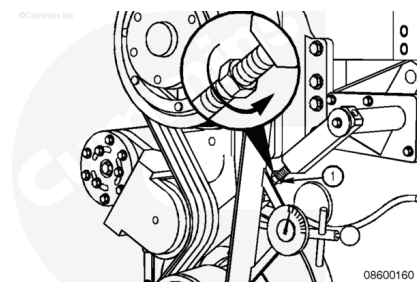
For the spring type tensioner **only**:

Note : The lower jam-nut has left-hand threads.

Loosen the jam nuts on the adjusting screw.

Note : The fan belt **must** be installed and under the tension of the fan idler arm spring to adjust the control rod. The fan belt and a portion of the flat washer are **not** shown for clarity.

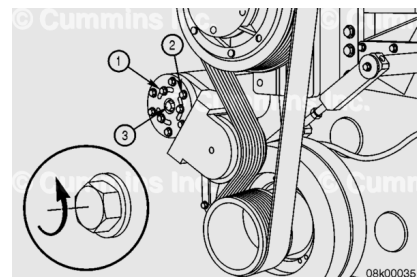
Turn the control rod (1) **clockwise** to tighten the belt tension.



For the spring type tensioner **only**:

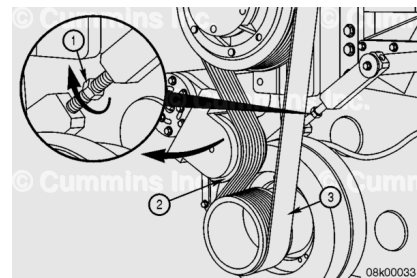
Note : Reposition the idler lever clamping screws **only** if correct tension can **not** be obtained using the control rod.

Loosen the three capscrews (1), the six capscrews (2), and the center capscrew (3) on the belt tensioner.



For the spring type tensioner **only**:

Turn the control rod (1) **counterclockwise** until the idler lever pulley (2) is off the belt (3) completely.



For the spring type tensioner **only**:

Position the idler lever pulley (2) against the belt (3).

Reposition the clamping capscrews if necessary.

Install the belt tension gauge, Part Number 3163569, onto the belt.

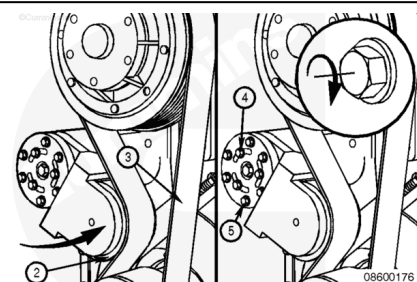
Turn the control rod **clockwise** to increase the belt tension.

Use the belt tensioner to set the belt initially to 4450 N [1000 lbf].

Operate the engine for a minimum of 1 hour to a maximum of 24 hours. Check and reset, if necessary, to 4450 N [1000 lbf].

After the belt tension has relaxed to a minimum of 3559 N [800 lbf], reset it to 4450 N [1000 lbf].

Tighten the three capscrews (4) and then the six capscrews (5) on belt tensioner.



Torque Value: 200 n•m [148 ft-lb]

For the slide type tensioner **only**:

Position the idler lever pulley (1) against the belt (2).

Install belt tension gauge, Part Number 3163569, onto the belt.

Reposition the clamping capscrews, if necessary.

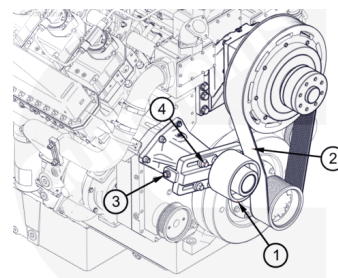
Turn the tensioning capscrew (3) clockwise to increase belt tension. Use the belt tensioner to set the belt initially to 4450 N [1000 lbf].

Operate the engine for a minimum of 1 hour to a maximum of 24 hours. Check and reset, if necessary, to 4450 N [1000 lbf].

After the belt tension has relaxed to a minimum of 3559 N [800 lbf], reset it to 4450 N [1000 lbf].

Tighten the four capscrews (4).

Torque Value: 280 n•m [207 ft-lb]



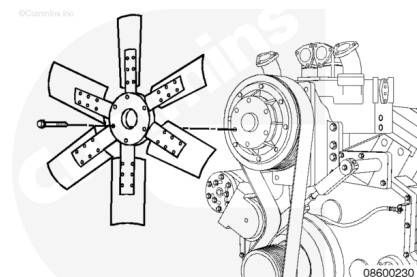
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Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the fan. Refer to Procedure 008-040 in Section 8. (</qs3/pubsys2/xml/en/procedures/89/89-008-040.html>)



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